

Installing VS Code and Wokwi Simulator on Windows

A step-by-step guide for setting up Visual Studio Code and the Wokwi simulation extension for embedded systems development.

Part 1: Installing Visual Studio Code

Step 1: Download VS Code

1. Open a web browser and go to **<https://code.visualstudio.com>**
2. Click the **Download for Windows** button (the site auto-detects your OS).
3. The installer file (e.g., VSCodeUserSetup-x64-x.xx.x.exe) will download to your **Downloads** folder.

Step 2: Run the Installer

1. Double-click the downloaded .exe file to launch the setup wizard.
2. Accept the license agreement and click **Next**.
3. Choose an install location (default is fine) and click **Next**.
4. On the **Select Additional Tasks** screen, it's recommended to check:
 - ✓ Add "Open with Code" action to Windows Explorer file context menu
 - ✓ Add "Open with Code" action to Windows Explorer directory context menu
 - ✓ Add to PATH (important — allows launching VS Code from terminal using code)
5. Click **Next**, then **Install**.
6. Once complete, click **Finish** to launch VS Code.

Step 3: Verify Installation

1. Open **Command Prompt** or **PowerShell**.
 2. Type:

```
code --version
```
 3. If it prints a version number, VS Code is installed correctly and added to PATH.
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Part 2: Installing the Wokwi Simulator Extension

Wokwi lets you simulate embedded hardware (ESP32, STM32, Arduino, etc.) directly inside VS Code, without needing physical boards.

Step 1: Open the Extensions Panel

1. Launch VS Code.
2. Click the **Extensions** icon in the left sidebar (or press

Ctrl+Shift+X).

Step 2: Search and Install Wokwi

1. In the Extensions search bar, type **Wokwi for VS Code**.
2. Select the extension published by **Wokwi**.
3. Click **Install**.

Step 3: Get a Wokwi License (Required for Simulation)

1. After installing, open the **Command Palette** (Ctrl+Shift+P).
2. Type and select **Wokwi: Request a New License**.
3. This opens a browser page — sign in and generate a free license key.
4. Copy the license key.
5. Back in VS Code, open the Command Palette again and select **Wokwi: Activate License** (or paste the key into the prompt shown after requesting it).

Step 4: Install Supporting Toolchains (for STM32/ESP32 projects)

Depending on what you're simulating, you may also need: - **PlatformIO IDE** extension (search in Extensions panel) — for building and flashing STM32/ESP32 firmware - **C/C++ Extension Pack** (by Microsoft) — for IntelliSense and debugging support

Step 5: Test Your Setup

1. Open or create a project folder containing:
 - Your firmware source code (.c/.cpp or .ino)
 - A wokwi.toml file (points Wokwi to your compiled firmware/binary)
 - A diagram.json file (defines the simulated circuit — boards, sensors, wiring)
2. Open the Command Palette and select **Wokwi: Start Simulator**.
3. A simulation panel should open showing your virtual circuit running.

Quick Reference

Task	Shortcut / Command
Open Extensions panel	Ctrl+Shift+X
Open Command Palette	Ctrl+Shift+P
Start Wokwi Simulation	Wokwi: Start Simulator (via Command Palette)
Request Wokwi License	Wokwi: Request a New License

Troubleshooting Tips

- **“code” not recognized in terminal:** Reinstall VS Code and ensure “Add to PATH” is checked, or manually add VS Code's bin folder to your system PATH.
- **Wokwi simulator won't start:** Confirm wokwi.toml and diagram.json exist in your project root and your license is

activated.

- **Firmware won't build:** Ensure PlatformIO or the relevant toolchain (e.g., STM32CubeIDE build tools) is installed and configured for your target board.